Competency Level 9.8: Uses control structures in developing programs

Learning Outcomes:

- Briefly describes control structures
- Lists and briefly describes the types of control structures
- Uses control structures appropriately in programming
- Applies nested control structures in programs

Control Structures

The flow of Control:
The flow of control in a program is implemented with three basic types of control structures:

- **Sequential**: Code statements are executed sequentially. (One line after another)
- **Selection**: Used to specify one or more conditions to be tested by the program.
- Python programming language provides the following decision-making statements:
    - if
    - if-else
    - if-elif-else
- **Repetition**: Used to execute a statement or group of statements multiple times. In Python, the following loops are used for repetition:
    - while
    - for

Indentation
To define ascope of a statement in the program, Python programming language uses indentation. Other programming languages often use curly-brackets for this purpose.

```
if b>a:
    print ("b greater than a")
```

Indentation

Example 11:
If statement:

Code **Output**

```
File Edit Format Run Options Windows Help
x = int(input("Enter your  ICT marks :"))
if x>=75:
        print("You are qualified for scholarship")
```

```
>>>
Enter your  ICT marks :85
You are qualified for scholarship
>>>
```

Python language provides multi-way conditional decision statements.

- **If-else Statement**
- **if, elif, else**

Example 12:
If-else Statement

Code

```
File  Edit  Format  Run  Options  Windows  Help
x = int(input("Enter your ICT marks :"))
if x >= 75:
                print("You are qualified for scholarship")
else:
                print("You are not qualified")
```

Output

```
>>>
Enter your ICT marks :58
You are not qualified
>>>
```

Example 13:
if, elif, else
Code

```
File  Edit  Format  Run  Options  Windows  Help
a = float(input("Enter your z-score : "))
if a>=2.65:
          print("you are eligible to study computer science")
elif a>=2.40:
          print("you are eligible to study information systems")
else:
     print("you are eligible to study ICT")
```

Output

```
>>>
Enter your z-score : 2.32
you are eligible to study ICT
>>>
```

Loops in Python

- while statement
- for statement

While loops:

```
while condition :
    block
```

Example 14:
The following program prints numbers from 1 to 5.

Code

```
File  Edit  Format  Run  Options  Windows  Help
i = 1            # initialization
while i < 6:     # condition
  print(i)       # display
  i += 1         # increment
```

Output

```
>>>
1
2
3
4
5
>>>
```

Python for loop:

A forloop is used to repeat a block of statements for a known number of times.

Syntex

for ***variable name*** in structure:

suit

Example 15:

Code Output

```
File Edit Format Run Options Windows Help
animals = ["cat", "dog", "fish"]
for a in animals:
  print(a)
```

```
>>>
cat
dog
fish
>>>
```

Variable **Sui** **Structure**

The break statement

With the breakstatement we can stop the loop even if the while condition is true:

Example 16:

Exit the loop when i is 3:

CodeOutput

```
ex17.py - C:\Python33\python program\Pyhon programn
File Edit Format Run Options Windows Help
x = 1
while x < 6:
  print(x,end=' ')
  if (x == 3):
    break
  x += 1
```

```
>>>
1 2 3
>>>
```

With the break statement we can stop the loop before it has looped through all the items:

Example 17:

Code Output

```
File Edit Format Run Options Windows Help
animals = ["cat", "dog", "fish"]
for i in animals:
  if i == "dog":
    break
  print(i)
```

```
>>>
cat
>>>
```

The Continue Statement
The current iteration stops with the continue statement and continues with the next:

Example 18:
Continue to the next iteration

if i is 3:

Code Output

```
File Edit Format Run Options
a = 0
while a < 6:
  a += 1
  if a == 3:
    continue
  print(a)
```

```
>>>
1
2
4
5
6
>>>
```

Example 19:
Code Output

```
File Edit Format Run Options Windows Help
animals = ["cat", "dog", "fish"]
for i in animals:
  if i == "dog":
    continue
  print(i)
```

```
>>>
cat
fish
>>>
```